



INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Course: BVWS103-OWASP Top 10 Vulnerabilities And
Exploitation Techniques(Security Misconfiguration)

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Title: Lab 7- Security Misconfiguration

Introduction

Security misconfiguration happens when an application is not properly set up to protect sensitive files and settings. Attackers can exploit this weakness to access confidential information such as database credentials. In this lab, I demonstrated how misconfiguration can be exploited and how it can be fixed.

Task 1: Exploiting Security Misconfiguration

Lab Setup

I created a vulnerable web application using PHP inline with the lecturer's guide. The application contained an `index.php` file, a `.htaccess` file, and a `config.php` file. The `config.php` file stored sensitive database credentials and was placed inside the public web directory. The files were stored in a folder called `Lab7_SecurityMisconfiguration`

```
kali@kali: /opt/lampp/htdocs
Session Actions Edit View Help

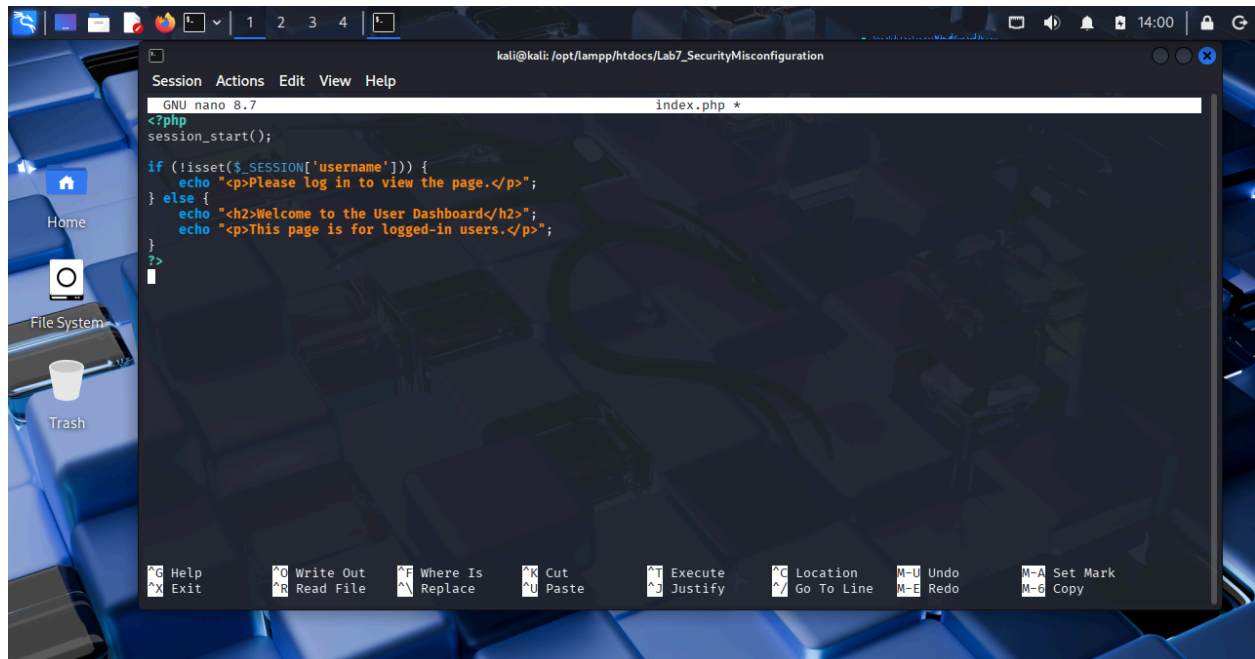
(kali@kali)-[~]
$ cd /opt/lampp/htdocs
(kali@kali)-[/opt/lampp/htdocs]
$ mkdir Lab7_SecurityMisconfiguration
(kali@kali)-[/opt/lampp/htdocs]
$ cd Lab7_SecurityMisconfiguration
(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ cd ..
(kali@kali)-[/opt/lampp/htdocs]
$ ls
applications.html  favicon.ico  lab1  lab4  Lab7_SecurityMisconfiguration  webalizer
bitnami.css       img         lab2  lab5_BrokenAccessControl  README.md  xss_form.php
dashboard         index.php   lab3  lab6  test.php

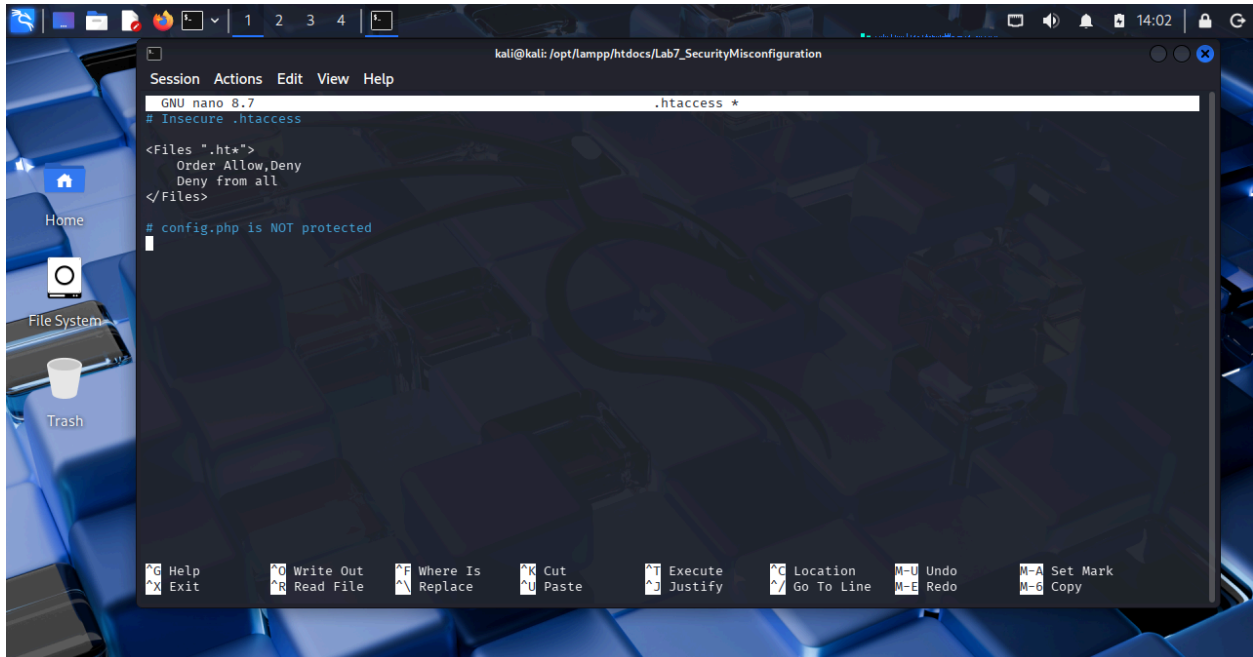
(kali@kali)-[/opt/lampp/htdocs]
$
```

```
kali@kali: /opt/lampp/htdocs/Lab7_SecurityMisconfiguration
Session Actions Edit View Help

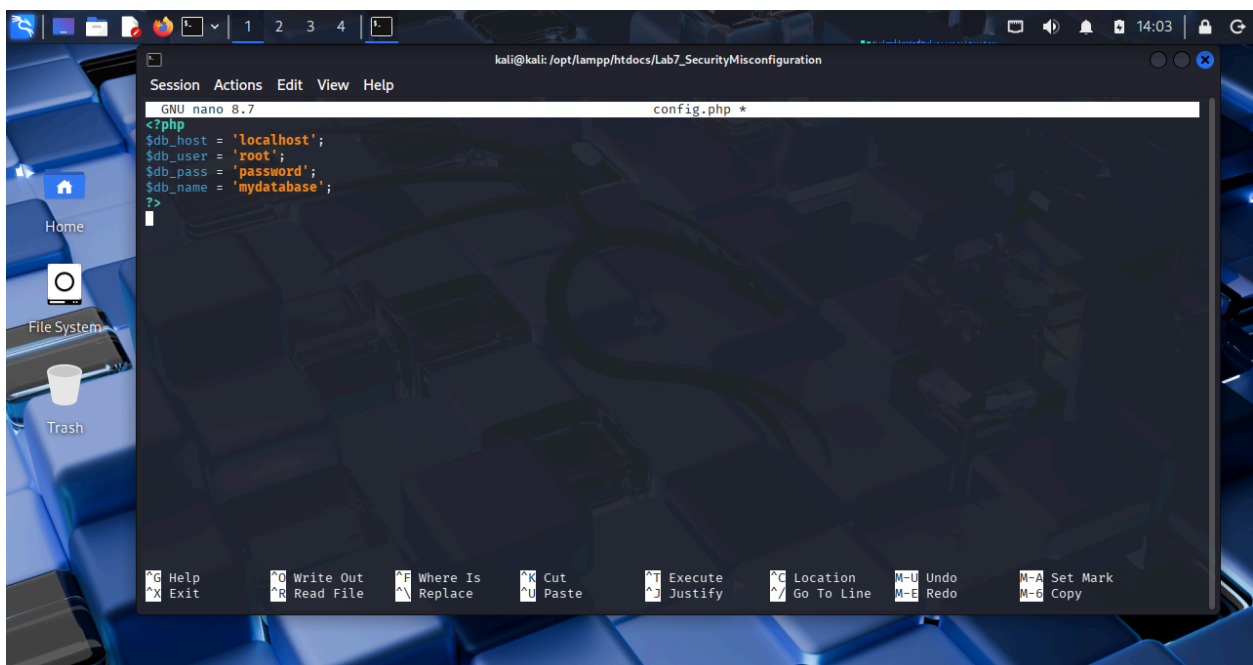
(kali@kali)-[/opt/lampp/htdocs]
$ nano index.php
(kali@kali)-[/opt/lampp/htdocs]
$ cd Lab7_SecurityMisconfiguration
(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ touch index.php .htaccess config.php
(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ ls
config.php  index.php
(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ ls -la
total 8
drwxrwxr-x 2 kali kali 4096 Feb 1 13:51 .
drwxrwxrwx 12 root root 4096 Feb 1 13:47 ..
-rw-rw-r-- 1 kali kali 0 Feb 1 13:51 config.php
-rw-rw-r-- 1 kali kali 0 Feb 1 13:51 .htaccess
-rw-rw-r-- 1 kali kali 0 Feb 1 13:51 index.php

(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$
```



A screenshot of a Kali Linux desktop environment. The background is a blue keyboard. On the left, there are icons for Home, File System, and Trash. A terminal window is open, showing the nano 8.7 editor editing the .htaccess file. The file content is:

```
# Insecure .htaccess<Files ".ht*">Order Allow,DenyDeny from all</Files># config.php is NOT protected
```

 The nano editor's status bar at the bottom shows various keyboard shortcuts like Help, Exit, Write Out, Read File, etc. The top of the terminal window shows the path: kali@kali: /opt/lampp/htdocs/Lab7_SecurityMisconfiguration.A screenshot of a Kali Linux desktop environment, similar to the one above. The nano 8.7 editor is now editing the config.php file. The file content is:

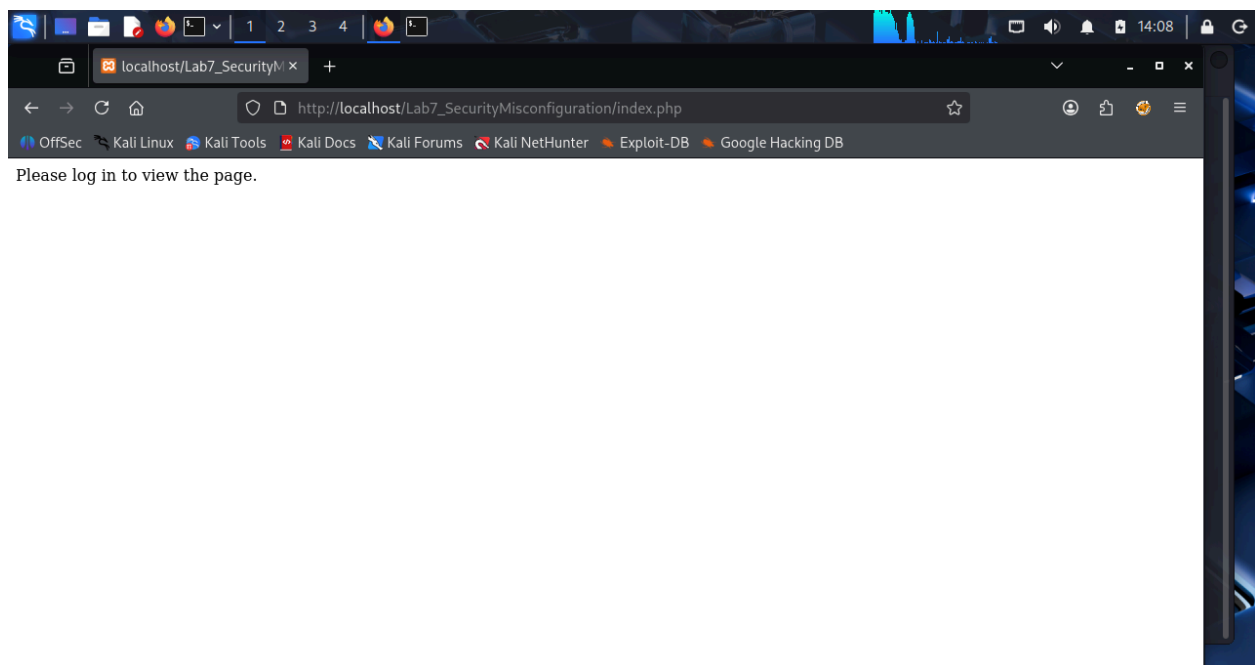
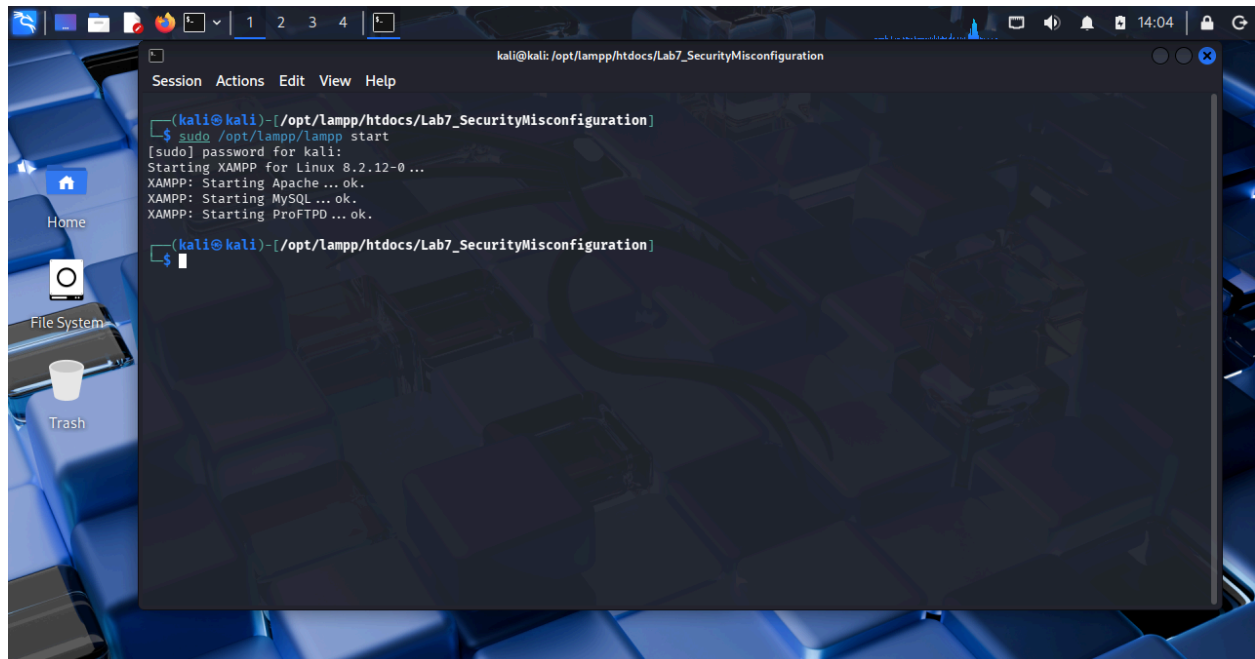
```
<?php$db_host = 'localhost';$db_user = 'root';$db_pass = 'password';$db_name = 'mydatabase';?>
```

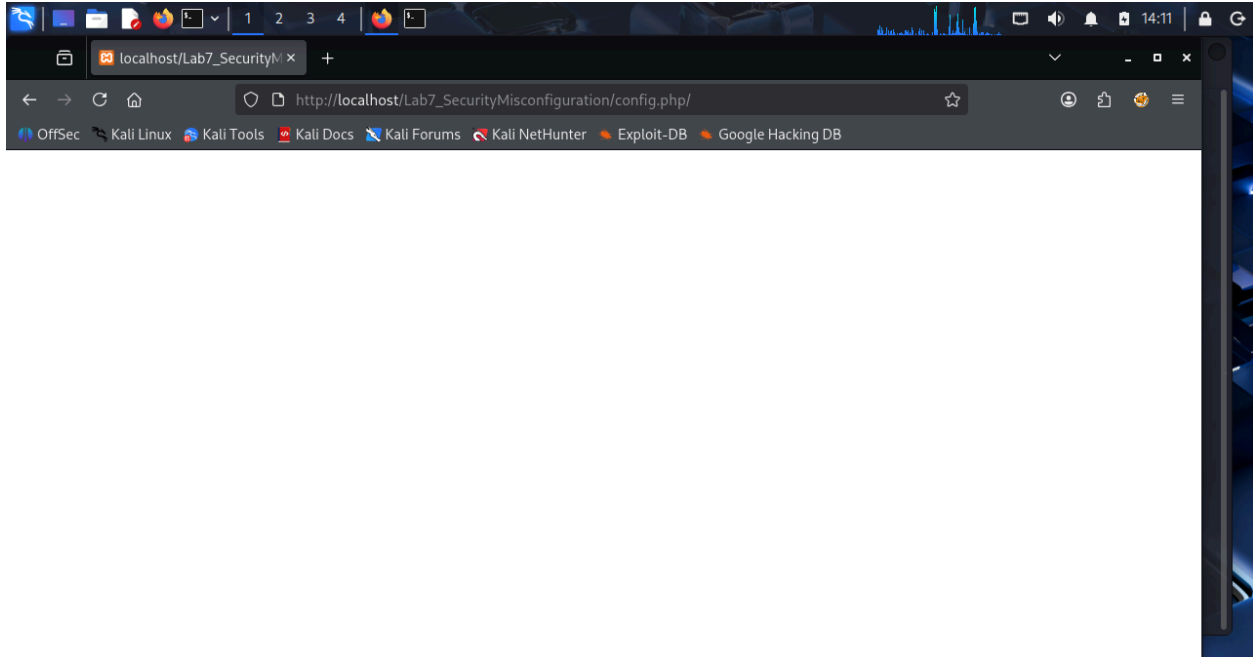
 The nano editor's status bar and the terminal window's path are the same as in the previous screenshot.

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Vulnerability Exploitation

The .htaccess file was improperly configured and did not restrict access to sensitive files. I accessed the config.php file directly through the browser. The file tried to display database credentials in plain text but it came blank, proving that sensitive information was publicly exposed.





Impact

An attacker could use this information to:

- Gain unauthorized database access
- Modify or delete data
- Compromise the entire web application

Task 2: Mitigating Security Misconfiguration

Securing Configuration Files

To fix the issue, I updated the `.htaccess` file to block access to sensitive files such as `.htaccess` and `config.php`. I also moved the `config.php` file outside the public web root to prevent direct access from the browser.

```
kali@kali: /opt/lampp/htdocs/Lab7_SecurityMisconfiguration
Session Actions Edit View Help
GNU nano 8.7 .htaccess *
# Secure .htaccess
<Files ".ht*">
  Order Allow,Deny
  Deny from all
</Files>

<Files "config.php">
  Order Allow,Deny
  Deny from all
</Files>
^
```

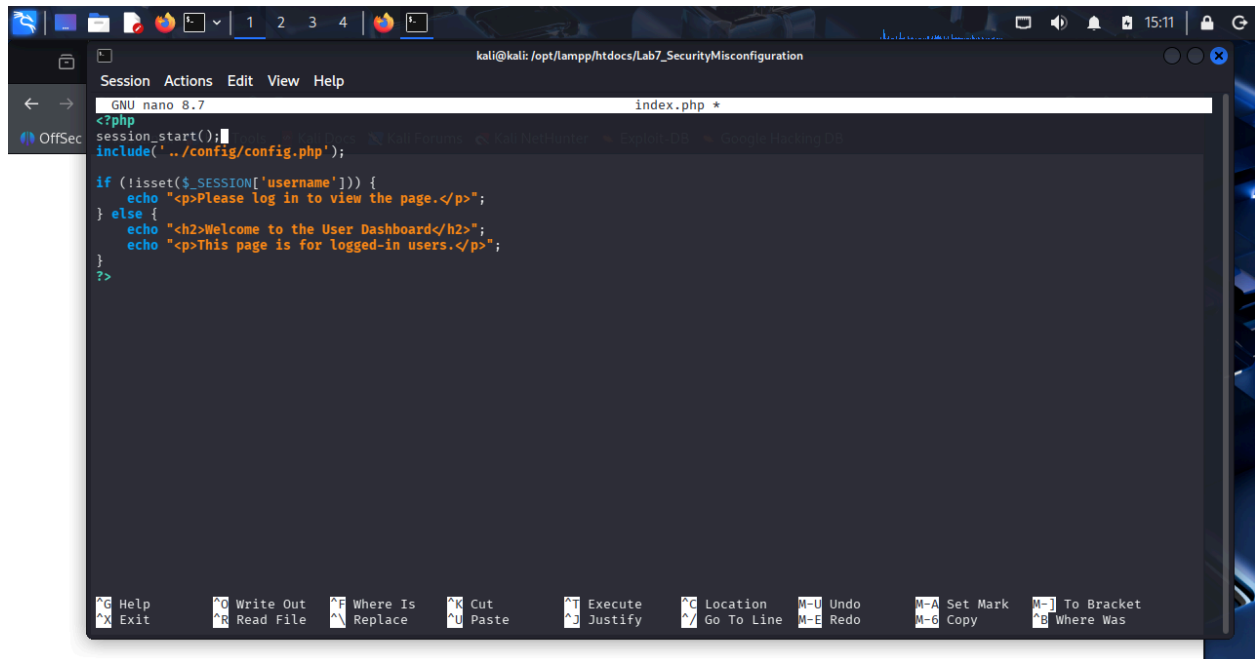
```
kali@kali: /opt/lampp/htdocs/Lab7_SecurityMisconfiguration
Session Actions Edit View Help
(kali@kali)-[/opt/lampp]
$ cd /htdocs/Lab7_SecurityMisconfiguration
cd: no such file or directory: /htdocs/Lab7_SecurityMisconfiguration

(kali@kali)-[/opt/lampp]
$ cd htdocs
(kali@kali)-[/opt/lampp/htdocs]
$ cd Lab7_SecurityMisconfiguration

(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ ls
config.php  index.php
(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ cat config.php
<?php
$db_host = 'localhost';
$db_user = 'root';
$db_pass = 'password';
$db_name = 'mydatabase';
?>

(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$ sudo mv config.php /opt/lampp/config/

(kali@kali)-[/opt/lampp/htdocs/Lab7_SecurityMisconfiguration]
$
```

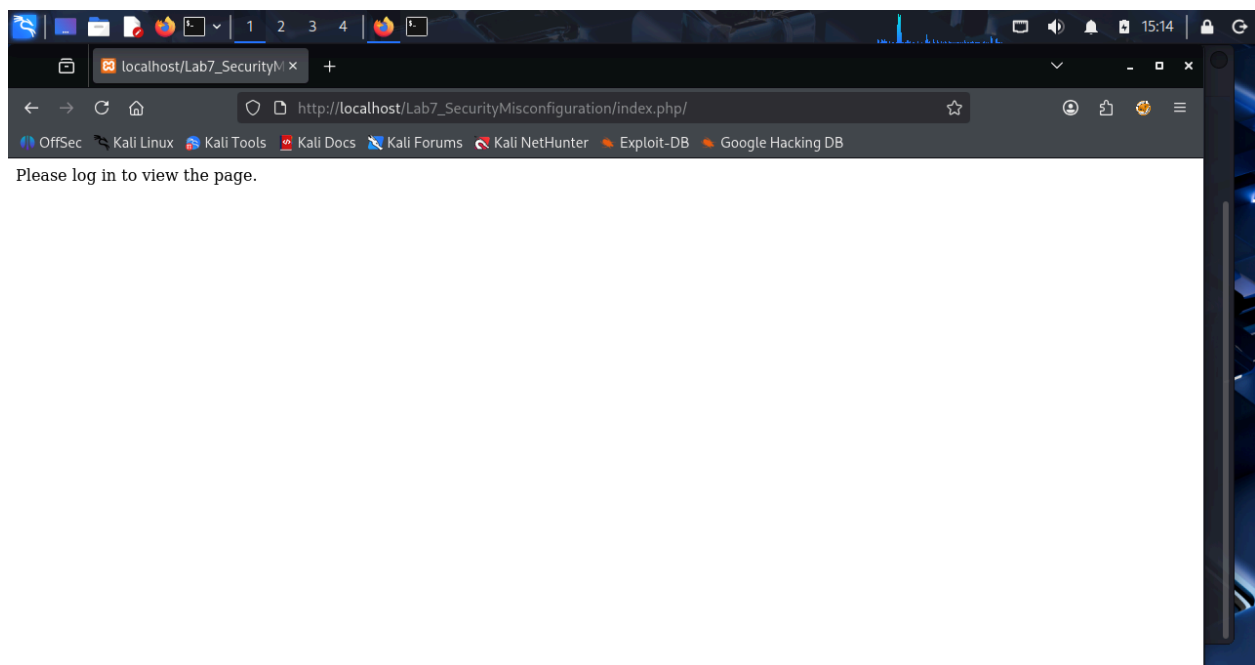
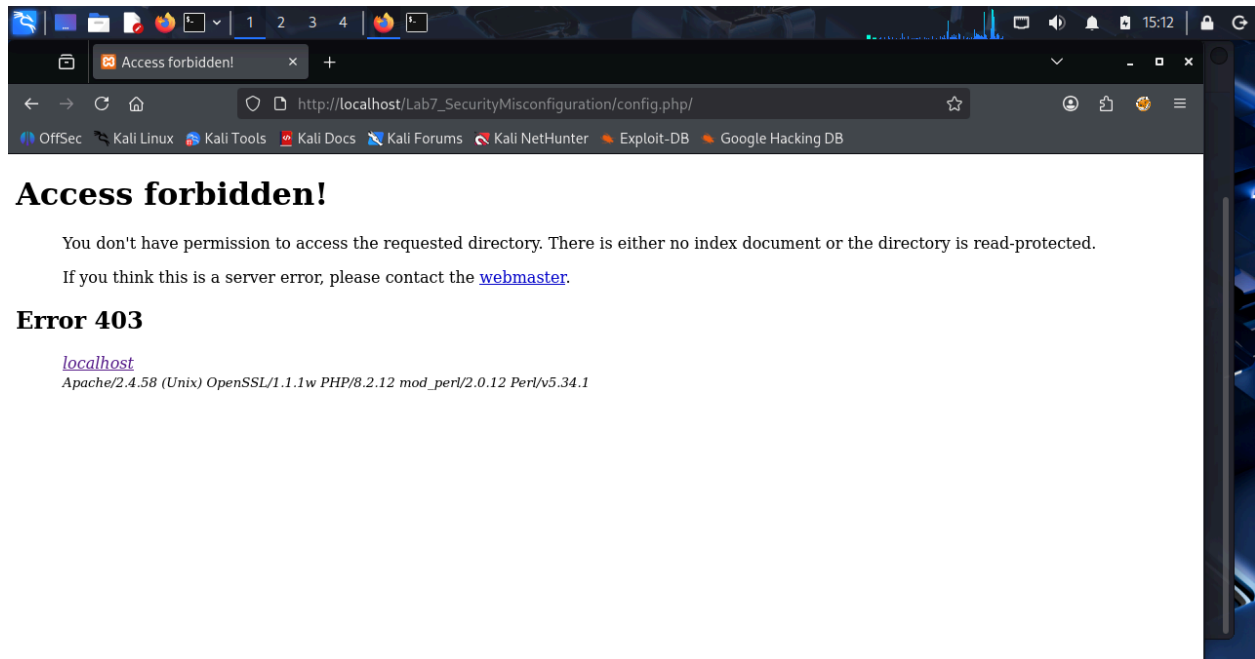



```
kali@kali: /opt/lampp/htdocs/Lab7_SecurityMisconfiguration
Session Actions Edit View Help
GNU nano 8.7 index.php *
<?php
session_start();
include('../config/config.php');

if (!isset($_SESSION['username'])) {
    echo "<p>Please log in to view the page.</p>";
} else {
    echo "<h2>Welcome to the User Dashboard</h2>";
    echo "<p>This page is for logged-in users.</p>";
}
?>
```

Testing the Fix

After applying the security controls, I tried accessing `config.php` through the browser again. The server returned a “403 Forbidden” error, confirming that the vulnerability had been successfully mitigated. `index.php` opened after refreshing.



Conclusion

This lab showed how poor configuration can expose sensitive information. By properly securing configuration files, restricting access, and moving sensitive files outside the public directory, the risk of exploitation is significantly reduced. Proper configuration management is essential for maintaining web application security.